

GAMMA MIA

①

Setting (X, d) metric space (can be topological, things don't change)

Functional $F: X \rightarrow \bar{\mathbb{R}}$

Problem: find $\min\{F(u): u \in X\}$

What? Γ -convergence, a notion of variational convergence for functionals
Introduced by De Giorgi in papers between 1975 and 1983

$F_\varepsilon: X_\varepsilon \rightarrow \bar{\mathbb{R}}$
 $F_0: X_0 \rightarrow \bar{\mathbb{R}}$ } informally, if $F_\varepsilon \xrightarrow{\Gamma} F_0 \Rightarrow \min\{F_\varepsilon(u): u \in X_\varepsilon\} \rightarrow \min\{F_0(u): u \in X_0\}$

Why? • Usually explicit solutions to minimization problems are not available (2)
 $\Rightarrow$ need for simulations or approximations, such that
minimizers to approximate energies approximate minimizers to exact energies
• Sometimes a physical system depends on parameters (small)
What's the asymptotic behaviour of the system? Is it modelled by a functional?

Really why? "Pointwise $< \Gamma < \text{Uniform}$ " (Weon somewhere else)

Take $f_n(x) = \cos(nx) : [0, 2\pi] \rightarrow \bar{\mathbb{R}}$ (or $f_\varepsilon(x) = \cos(\frac{x}{\varepsilon})$)

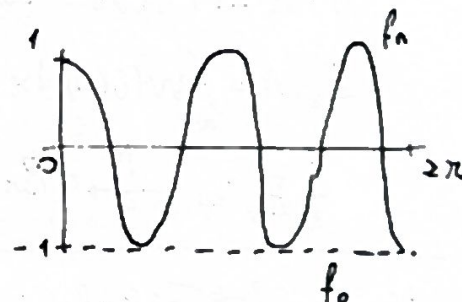
$$\min\{f_n(x) : x \in [0, 2\pi]\} = -1$$

Pointwise limit? Does not exist

Uniform limit? Does not exist

Weon limit? $f_w(x) \equiv 0$ but $\min\{f_w(x) : x \in [0, 2\pi]\} = 0 \neq -1$

Turns out $\Gamma\text{-}\lim_{n \rightarrow \infty} f_n = f_0 \equiv -1$



Examples • Mumford-Shah for image segmentation

Maybe do there at the end

$$MS(u) = \int_{\Omega} |\nabla u|^2 dx + H^{n-1}(\mathcal{J}_u) \quad u \in SBV(\Omega)$$

• Ambrosio-Tortorelli approximation with phase field

$$AT_\varepsilon(u, v) = \int_{\Omega} (\eta_\varepsilon + v^2) |\nabla u|^2 dx + \int_{\Omega} (\varepsilon |\nabla v|^2 + \frac{(v-1)^2}{9\varepsilon}) dx \quad (u, v) \in H^1(\Omega) \times (H^1(\Omega) \cap L^\infty(\Omega))$$

$(u_\varepsilon, v_\varepsilon) \rightarrow (u_0, 1)$ where u_0 minimizes $MS(u)$

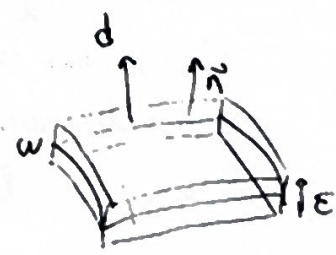
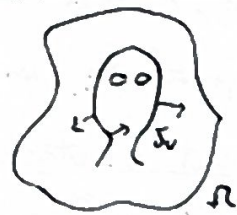
• Dimensional reduction

2D $\text{microscale } u \in \mathbb{R}^2$ and $\Omega_\varepsilon^\sim = \omega \times [-\frac{1}{2}, \frac{1}{2}]$

$$F_\varepsilon(u) = \int_{\Omega_\varepsilon^\sim} W_\varepsilon(\nabla u) dx \xrightarrow{\Gamma} F_0(u) = \int_{\omega} \bar{W}(\nabla u, d) d\mathcal{H}^{n-2}(x)$$

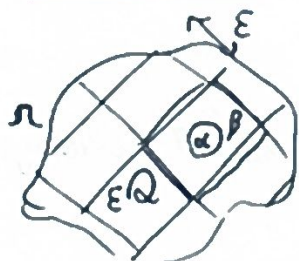
d determined by microstructure and/or curvature $\bar{W}(\nabla u) = QW_0(\nabla' u)$

$\nabla' u = (\partial_1 u, \partial_2 u)$, $W_0(F) = \max_{\theta \in \mathbb{S}^{n-2}} W(F \otimes \theta)$, QW_0 quasiconvex envelope



⑦

• Homogenization



$\varrho: \mathbb{R}^n \rightarrow \{1, \rho\}$ Q -periodic, property of material

$$E_\varepsilon(u) = \frac{1}{2} \int_\Omega \varrho\left(\frac{x}{\varepsilon}\right) |\nabla u|^2 dx = \int_\Omega f_\varepsilon\left(\frac{x}{\varepsilon}, \nabla u\right) dx$$

$$E_\varepsilon \xrightarrow{\Gamma} E_0(u) = \int_\Omega f_{\text{hom}}(\nabla u) dx \quad f_{\text{hom}}(z) := \lim_{T \rightarrow \infty} \frac{1}{T^n} \int_{TQ} \varrho\left(\frac{x}{T}\right) |\nabla u|^2 dx$$

$$\lim_{\varepsilon \rightarrow 0} \frac{1}{2} \int_\Omega \varrho\left(\frac{x}{\varepsilon}\right) |\nabla u|^2 dx \xrightarrow{\Gamma} \frac{1}{2} \int_\Omega \varrho_{\text{hom}} |\nabla u|^2 dx$$

$$f_{\text{hom}}(z) = \lim_{T \rightarrow \infty} \min \left\{ \frac{1}{T^n} \int_{TQ} f(s, z, \nabla v) ds : v \in W_0^{1,p}(TQ) \right\} \quad \text{where } \varrho_{\text{hom}} := \frac{1}{\int_0^1 \frac{1}{\varrho(x)} dx}$$

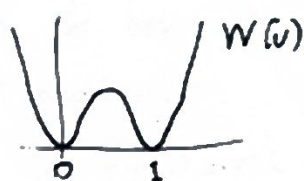
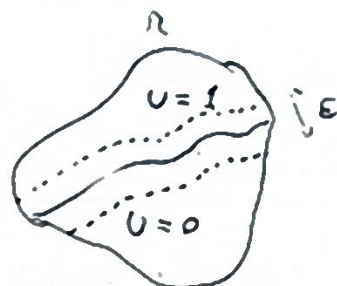
• Phase separation

Ω filled with immiscible, incompressible fluids

$u: \Omega \rightarrow \{0, 1\}$ divides the fluid

$$\text{Gurtin 1977: } E(u) = \int_\Omega W(u) dx + \int_\Omega u dx = c$$

with $c = \frac{|\Omega|}{2}$ in example, too many solutions



Modica 1981: penalize interface

$u: \Omega \rightarrow \{0, 1\}$: fluids can mix close to interface (size ε)

$$E_\varepsilon(u) = \int_\Omega W(u(x)) dx + \varepsilon^2 \int_\Omega |\nabla u(x)|^2 dx$$

$$\frac{1}{\varepsilon} E_\varepsilon(u) \xrightarrow{\Gamma} \sigma \text{Per}(\{u=0\}; \Omega) \quad u \in BV(\Omega; \{0, 1\})$$

~~Blueballing~~ but: Preliminaries: Start with Weierstrass

• Lower semicontinuity (lsc)

$f: X \rightarrow \overline{\mathbb{R}}$ is lsc if $\{x \in X : f(x) \leq t\}$ is closed for every $t \in \mathbb{R}$

Equivalently, in metric spaces $f(x) \leq \liminf_{n \rightarrow \infty} f(x_n) \quad \forall x_n \rightarrow x$

• Direct method

$F: X \rightarrow \mathbb{R}$: when does $\min\{F(u) : u \in X\}$ exist?

1) $K \subseteq \mathbb{R}^n$ compact, $F: K \rightarrow \mathbb{R}$ continuous $\Rightarrow \exists \min$

More in f always exists $\Rightarrow$ tone infimizing sequence (u_n) :

$$F(u_n) \rightarrow \inf_K F$$

Then (u_n) sequence in a compact $\Rightarrow$ has limit u_0

$$F \text{ continuous} \Rightarrow F(u_n) \rightarrow F(u_0) = \inf_K F$$

2) $F: K \rightarrow \mathbb{R}$ lsc

$$F(u_0) = \liminf_{n \rightarrow \infty} F(u_n) = \lim_n F(u_n) = \inf_K F$$

3) No compactness $\Rightarrow F$ coercive ($\{u : F(u) \leq t\}$ bounded $\forall t \in \mathbb{R}$)

Then (u_n) minimizing sequence

$$\Rightarrow F(u_n) \leq t = \inf_X F + 1 \quad \text{for } n \text{ large enough}$$

coercive $\Rightarrow \{F(u_n) \leq t\}$ bounded } compact $\Rightarrow \exists u_{n_n} \rightarrow u_0$
 l.s.c. $\Rightarrow$ closed

Arriving at the definition

(4)

We need: convergence of minima and minima points

stable under continuous perturbation ($\Gamma\text{-}\lim(F_\varepsilon + G) = \Gamma\text{-}\lim(F_\varepsilon) + G$)

$F_\varepsilon : X_\varepsilon \rightarrow \bar{\mathbb{R}}$ } We call (u_ε) a minimizing sequence if
 $F_0 : X_0 \rightarrow \bar{\mathbb{R}}$ }

$$\lim_{\varepsilon \rightarrow 0} [F_\varepsilon(u_\varepsilon) - \inf_{X_\varepsilon} F_\varepsilon] = 0$$

WLOG suppose $X_\varepsilon \subseteq X_0 \quad \forall \varepsilon$ and extend F_ε to $\tilde{F}_\varepsilon(u) = \begin{cases} F_\varepsilon(u) & u \in X_\varepsilon \\ +\infty & u \in X_0 \setminus X_\varepsilon \end{cases}$
 For (u_ε) to have a limit u_0 we need compactness, upper and lower bounds

• compactness

We require every F_ε to be coercive

$\Rightarrow \{F_\varepsilon\}$ equicoercive

$$\forall t \in \mathbb{R} \quad \exists K_t \text{ compact} : \{F_\varepsilon \leq t\} \subseteq K_t \quad \Rightarrow u_\varepsilon \rightarrow u_0$$

• upper bound on $\inf_X F_\varepsilon$

$$\limsup_{\varepsilon \rightarrow 0} (\inf_X F_\varepsilon) = \lim_{\varepsilon \rightarrow 0} (\sup_{m \geq \varepsilon} \inf_X F_m) = \inf_X \sup_{m \geq \varepsilon} \inf_X F_m \leq \inf_X F = F(u_0)$$

We can see that this holds $\forall u \in X : \limsup_{\varepsilon \rightarrow 0} (\inf_X F_\varepsilon) \leq F(u)$

Equivalent to: $\forall u \in X \quad \exists (u_\varepsilon) : u_\varepsilon \rightarrow u : \limsup_{\varepsilon \rightarrow 0} F_\varepsilon(u_\varepsilon) \leq F(u)$

• lower bound on $\inf_X F_\varepsilon$

$$(\inf_X F) \leq F(u_0) \leq \liminf_{\varepsilon \rightarrow 0} F_\varepsilon(u_\varepsilon) \leq \liminf_{\varepsilon \rightarrow 0} (\inf_X F_\varepsilon)$$

$$\Rightarrow \forall u \in X, \forall (u_\varepsilon) : u_\varepsilon \rightarrow u : F(u) \leq \liminf_{\varepsilon \rightarrow 0} F_\varepsilon(u_\varepsilon)$$

Conclusion: If $u_\varepsilon \rightarrow u_0$ and you have the bounds

$$\limsup_{\varepsilon \rightarrow 0} (\inf_X F_\varepsilon) \leq \inf_X F \leq F(u_0) \leq \liminf_{\varepsilon \rightarrow 0} F_\varepsilon(u_\varepsilon) = \liminf_{\varepsilon \rightarrow 0} (\inf_X F_\varepsilon) (= \limsup_{\varepsilon \rightarrow 0} (\inf_X F_\varepsilon))$$

Therefore • $\inf_X F = \lim_{\varepsilon \rightarrow 0} \inf_X F_\varepsilon$

$$\bullet \min \{F(u) : u \in X\} = F(u_0)$$

$$\bullet u_\varepsilon \rightarrow u_0 \in \arg\min \{F(u) : u \in X\}$$

Definition: Γ -convergence

(X, d) metric space, $(F_j), F_0 : X \rightarrow \bar{\mathbb{R}}$

$$F_j \xrightarrow{\Gamma} F_0 \text{ if } \forall x \in X$$

$$i) \forall x_j \rightarrow x \text{ it holds } F(x) \leq \liminf_{j \rightarrow \infty} F_j(x_j)$$

$$ii) \exists \bar{x}_j \rightarrow x \text{ s.t. } F(x) \geq \limsup_{j \rightarrow \infty} F_j(\bar{x}_j)$$

$$\text{Equivalently } iii) \exists \bar{x}_j \rightarrow x \text{ s.t. } F(x) = \lim_{j \rightarrow \infty} F_j(\bar{x}_j) \quad (\text{recovery sequence})$$

Question: is the Γ -convergence induced by a norm?

(*) NO, as it can be seen from

$$F_n \equiv F \Rightarrow \min_j i \Rightarrow F(x) \leq \liminf_{j \rightarrow \infty} F_j(x_j) = \liminf_{j \rightarrow \infty} F(x_j)$$

only true if F is l.s.c.

it holds in particular that $\Gamma\text{-lim}_{j \rightarrow \infty} F = F^{\text{lsc}}(u) = \sup \{G(u) : G \text{ l.s.c., } G \leq F\}$

This is useful if F is not l.s.c., as $\inf_x F = \min_x F^{\text{lsc}}$

Theorem fundamental theorem of Γ -convergence

(X, d) metric space, $(F_j), F_0 : X \rightarrow \bar{\mathbb{R}}$ s.t. $F_j \xrightarrow{\Gamma} F_0$

and (F_j) is equi-mildly coercive (EMC)

$$\exists K \subseteq X \text{ compact : } \inf_x F_j = \inf_K F_j \quad \forall j$$

$$\Rightarrow 1) \exists \min_x F_0$$

$$2) \exists \lim_{j \rightarrow \infty} \inf_x F_j = \min_x F_0$$

3) if (u_j) is a precompact sequence s.t. $\lim_{j \rightarrow \infty} F_j(u_j) = \lim_{j \rightarrow \infty} \inf_x F_j$
then every limit point of (u_j) is a minimum point for F_0

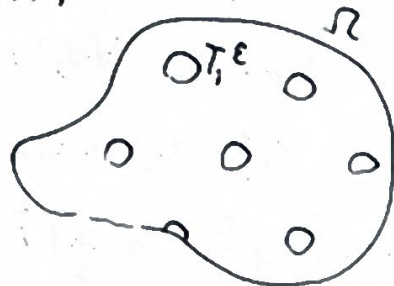
For (3) the opposite is not true: $f_j(x) = \frac{1}{j}|x|$ argmin $F_j = \{0\}$
 $f_j \xrightarrow{\Gamma} f_0 \equiv 0$ argmin $f_0 = \mathbb{R}^n$

check time
tolerant
asymptotic expansion

$\Rightarrow \Gamma$ -convergence also reflects the interwining minimum points.

Strange term coming from nowhere Ciarraesca-Murat 1997

$$\begin{cases} -\Delta u^\varepsilon = f & x \in \Omega^\varepsilon \\ u^\varepsilon = 0 & x \in \partial\Omega^\varepsilon \end{cases} \quad \Omega^\varepsilon = \Omega \setminus \bigcup_{i=1}^{n^\varepsilon} T_i^\varepsilon \subset \mathbb{R}^d$$



7.b.15

$$\Rightarrow \int_{\Omega^\varepsilon} \nabla u^\varepsilon \cdot \nabla v \, dx = \int_{\Omega} f v \, dx \quad \forall v \in H_0^1(\Omega^\varepsilon)$$

Denote $\tilde{u}^\varepsilon$ the extension of u^ε to $\tilde{u}^\varepsilon = \begin{cases} u^\varepsilon & \text{in } \Omega^\varepsilon \\ 0 & \text{in } T_i^\varepsilon \end{cases}$

$$\Rightarrow \tilde{u}^\varepsilon \in H_0^1(\Omega) \text{ and } \int_{\Omega} |\nabla \tilde{u}^\varepsilon|^2 \, dx \leq \|f\|_{L^2} \cdot \|\tilde{u}^\varepsilon\|_{L^2}$$

$$\Rightarrow \exists u \in H_0^1(\Omega) : \tilde{u}_\varepsilon \rightharpoonup u \text{ weakly}$$

Suppose the following: $\exists w_\varepsilon$ functions and $\exists \mu$ distribution s.t.

H1) $w^\varepsilon \in H^1(\Omega)$

H2) $w^\varepsilon = 0$ on T_i^ε

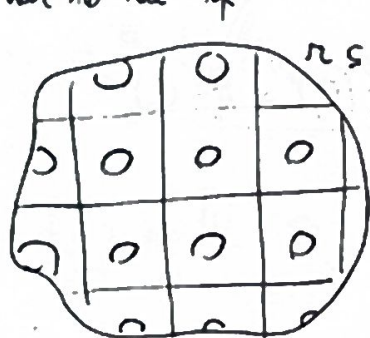
H3) $w^\varepsilon \rightarrow 1$ weakly in $H^1(\Omega)$

H4) $\mu \in W^{-1,\infty}(\Omega)$

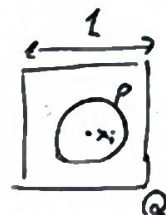
H5) $\forall (v^\varepsilon) : v^\varepsilon = 0$ on T_i^ε s.t. $v^\varepsilon \rightarrow v$ in $H^1(\Omega) \Rightarrow \langle -\Delta w^\varepsilon, \varphi v^\varepsilon \rangle_{H^1(\Omega), H^1(\Omega)} \rightarrow \langle \mu, \varphi v \rangle_{H^1(\Omega), H^1(\Omega)}$ $\forall \varphi \in D(\Omega)$

$\Rightarrow v$ is unique solution of
$$\begin{cases} -\Delta v + \mu v = f & \text{in } \Omega \\ v \in H_0^1(\Omega) \end{cases}$$

Easier to see if



$P_\varepsilon = \bigcup_{i=1}^{N_\varepsilon} P_{x_i}$ where $P_{x_i} = x_i + P$



$\Omega_\varepsilon = \Omega \setminus P_\varepsilon$

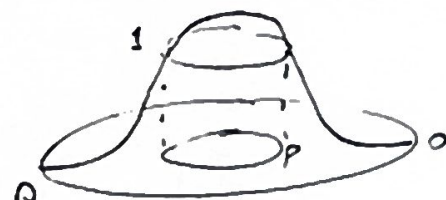
Consider
$$\begin{cases} -\Delta v^\varepsilon = f & \text{on } \Omega_\varepsilon \\ v^\varepsilon = 0 & \text{on } \partial \Omega_\varepsilon \end{cases}$$

$$\begin{cases} -\Delta v + \mu v = f & \text{in } \Omega \\ v = 0 & \text{in } \partial \Omega \end{cases}$$

or $\delta_\varepsilon \approx e^{-\varepsilon^{-2}}$

with $\alpha = 4\pi C_P(P, Q)$

$C_P(P, Q) = \int_Q |\nabla u|^2 dx : u \in C_0^\infty(Q), u \geq \mathbb{1}_P$

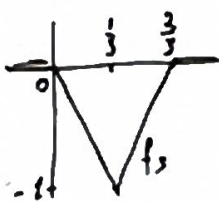


In $\mathbb{R}^N \Rightarrow \delta_\varepsilon \approx \varepsilon^{\frac{N}{N-2}}$

Examples

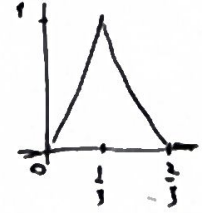
• $X = \mathbb{R} \quad f_\varepsilon(x) = \sin(\varepsilon x) \xrightarrow{P} f(x) = -1$

• $X = \mathbb{R} \quad f_\varepsilon(x) = \begin{cases} 0 & x \neq 0 \\ -1 & x = 0 \end{cases} \xrightarrow{P} \begin{cases} 0 & x \neq 0 \\ -1 & x = 0 \end{cases}$



recovery requires $x_\varepsilon = \frac{1}{\varepsilon} : f_\varepsilon(x_\varepsilon) = -1 \quad \forall \varepsilon$

• $X = \mathbb{R} \quad g_\varepsilon = -f_\varepsilon$



$g_\varepsilon \xrightarrow{P} 0$

$\Rightarrow P$ convergence is not symmetric

• $X = \mathbb{R} \quad h_\varepsilon(x) = \begin{cases} f_\varepsilon(x) & \text{even} \\ 2f_\varepsilon(x) & \text{odd} \end{cases} \Rightarrow P\text{-}\lim h_\varepsilon(x) = \begin{cases} 0 & x \neq 0 \\ \neq & x = 0 \end{cases}$

since $P\text{-}\liminf h_\varepsilon(0) = -2 \neq -1 = P\text{-}\limsup h_\varepsilon(0)$

• If $F_\varepsilon \rightarrow F$ pointwise $\Rightarrow P\text{-}\lim F_\varepsilon \leq F$

• If $F_\varepsilon \rightarrow F$ uniformly $\Rightarrow P\text{-}\lim F_\varepsilon = F^{esc}$

• $f_\varepsilon = \sin^2(\varepsilon x) \xrightarrow{P} 0$ but $g_\varepsilon = \cos^2(\varepsilon x) \xrightarrow{P} 1$

Proof of fundamental theorem of P -convergence

(5.515)

We'll prove only ① and ②

Using EML, $\Rightarrow$ we can find a sequence $(\tilde{\sigma}_j) \subset K$ s.t. $\liminf_{j \rightarrow \infty} F_j(\tilde{\sigma}_j) = \liminf_{j \rightarrow \infty} \inf_x F_j$

K compact $\Rightarrow \exists (\tilde{\sigma}_{j_n}) \subset K : \tilde{\sigma}_{j_n} \rightarrow u_0 \in K$

WLOG we consider $(\tilde{\sigma}_{j_n})$ along which $\liminf_{j \rightarrow \infty}$ is a limit

$$\Rightarrow \liminf_{j \rightarrow \infty} F_j(\tilde{\sigma}_j) = \lim_{n \rightarrow \infty} F_{j_n}(\tilde{\sigma}_{j_n})$$

$$\text{We define then } u_j := \begin{cases} \tilde{\sigma}_{j_n} & j = j_n \\ u_0 & j \neq j_n \end{cases} \Rightarrow u_j \rightarrow u_0$$

$$\text{Then } \inf_x F \leq F(u_0) \leq \liminf_{j \rightarrow \infty} F_j(u_j) \leq \liminf_{n \rightarrow \infty} F_{j_n}(\tilde{\sigma}_{j_n}) = \lim_{n \rightarrow \infty} F_{j_n}(\tilde{\sigma}_{j_n})$$

$$= \liminf_{j \rightarrow \infty} \inf_x F_j \leq \limsup_{j \rightarrow \infty} \inf_x F_j \leq \limsup_{j \rightarrow \infty} F_j(\tilde{\sigma}_j) \leq F(u_0)$$

by taking $\tilde{\sigma}_j \rightarrow u_0$ as a recovery sequence

then we conclude $\checkmark$